Polycystic ovary syndrome detection using neural network.

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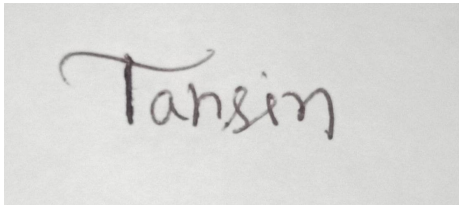
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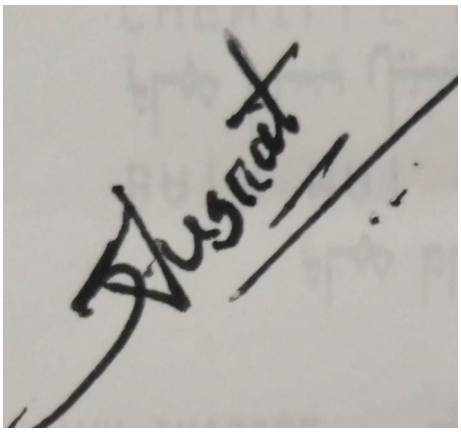
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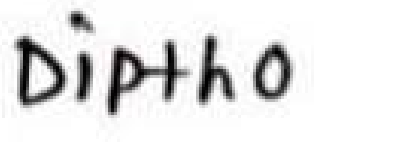
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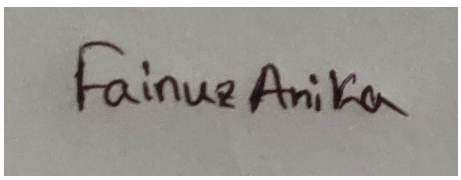
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Abstract

A fairly frequent endocrine abnormality among women of reproductive age is polycystic ovary syndrome (PCOS). In this disease, the ovaries produce abnormally high levels of androgens, which are male sex hormones that are typically present in women in trace amounts. The basic difference between the PCOS and normal ovarian cysts is the substantial hormonal imbalance which is not a general case in ovarian cysts. Study says, among 15 percentages of reproductive women this disease is found which is a major cause of women infertility. Even though this is a very common worldwide spreaded serious disease, it is hard to diagnose properly. So firstly, since this is a worldwide problem a lot of people are thinking but they could not come to a conclusion. Secondly, detecting this disorder is very difficult since the symptoms of PCOS match with other diseases which makes the detection difficult. For this reason, we basically got interested in this area.

Keywords: Otsu threshold; Machine Learning; Folicl̈es; KNN Algoritm; Linear Regression Analysis; Androgens;

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Chapter 1

Introduction

1.1 Introduction

Women of reproductive age are frequently affected by the endocrine condition polycystic ovarian syndrome (PCOS). It is characterized by hormonal abnormalities that can cause a range of symptoms, including irregular periods, acne, and excessive hair growth, as well as the existence of cysts in the ovaries. Additionally, PCOS can raise your risk of contracting diseases including diabetes and heart disease.Currently, PCOS is diagnosed using a mix of clinical and laboratory indicators, including irregular menstrual periods, high androgen hormone levels, and the presence of cystic ovaries on ultrasound. It has been demonstrated that ultrasound imaging, a non-invasive and frequently used technique for assessing the ovaries, is a valuable tool for recognizing the distinctive cystic abnormalities seen in PCOS.In this study, we want to look into how ultrasound imaging can be used to diagnose PCOS. In order to pinpoint specific traits that are connected to PCOS, ultrasound images of the ovaries will be examined. The objective of this study is to create a more precise and effective approach of detecting PCOS using ultrasound imaging, which may enhance the diagnosis procedure and result in earlier treatment for those who are affected. Additionally, as part of this study, we'll investigate how to evaluate ultrasound images and increase the precision of PCOS detection by using sophisticated image processing methods like machine learning algorithms. These methods can be trained to recognize patterns that could be challenging for human observers to notice, and they have the potential to automate the process of recognizing the distinctive cystic alterations seen in PCOS.Additionally, this study may help in the discovery of new PCOS biomarkers that can be utilized to speed up diagnosis. We want to find distinct patterns or traits that are specific to PCOS and can be utilized to distinguish it from other conditions by examining the ultrasound images of the ovaries in people with the condition.The overall goals of this study are to increase the precision and effectiveness of PCOS identification and to show the potential of ultrasound imaging as a diagnostic tool for PCOS. The findings of this study may have a big impact on how PCOS is managed and treated, which will eventually lead to better health outcomes for those who have it.

1.1.1 Problem Statement

A prevalent endocrine condition known as polycystic ovarian syndrome (PCOS) that affects women of early age is defined by the presence of several tiny cysts in the ovaries.Ultrasound imaging is a commonly used method for detecting PCOS, but the process can be time-consuming and require specialized training. The problem to be solved is to develop a computer-aided diagnostic system that can accurately and efficiently detect PCOS using ultrasound images, to improve the diagnostic process and ultimately improve patient outcomes.This problem statement aims to develop a computer-aided diagnostic system that can automatically detect PCOS using ultrasound images. The system should be able to analyze the images and identify the presence of cysts in the ovaries, while also considering other factors such as the size and shape of the ovaries. The system should be able to produce a diagnosis with a high level of accuracy, and should be easy to use for medical practitioners with little or no specialized training in ultrasound imaging. Additionally, the system should be able to process the images in a timely manner, to minimize the wait time for patients and to improve the efficiency of the diagnostic process. The ultimate goal of this problem statement is to improve the diagnosis and treatment of PCOS, by providing a reliable and efficient diagnostic tool that can be used by healthcare professionals.Furthermore, this problem statement also aims to address the issue of inter-observer variability, which is a common challenge in ultrasound-based diagnosis of PCOS. Inter-observer variability refers to the fact that different practitioners may have different levels of experience and expertise in interpreting ultrasound images, which can lead to inconsistent or inaccurate diagnoses. No matter how experienced the practitioner utilizing the computer-aided diagnostic system is, the system has to be able to deliver objective and consistent results. Additionally, this problem statement also considers the importance of patient privacy and data security. The system should be designed to protect patient privacy by adhering to relevant regulations and guidelines, and by implementing appropriate measures to secure the ultrasound images and patient data. In summary, the problem to be solved is to develop a computer-aided diagnostic system that can accurately and efficiently detect PCOS using ultrasound images, to improve the diagnostic process, minimize inter-observer variability, and ensure patient privacy and data security. This will ultimately improve patient outcomes and make the diagnostic process more efficient for healthcare professionals.

1.1.2 Research Objective

The objective of research on Polycystic ovary syndrome (PCOS) detection would likely be to develop or improve methods for accurately identifying and diagnosing the condition in individuals, with the goal of improving patient outcomes through earlier detection and treatment. This may involve studying biomarkers, symptoms, and risk factors associated with PCOS, as well as evaluating the effectiveness of different diagnostic techniques and tools. Ultimately, the goal would be to develop a reliable and efficient method of detecting PCOS that can be easily implemented in clinical settings.Additionally, research could also focus on understanding the underlying causes of PCOS, which is a complex and poorly understood condition. This could include studying the genetic and environmental factors that contribute to the development of PCOS, as well as investigating potential mechanisms of disease. An-

other important area of research could be the development of new treatment options for PCOS, as current therapies can have limited efficacy and significant side effects. This could include researching new medications or therapies that target specific mechanisms of the disease, as well as investigating the use of lifestyle interventions, such as diet and exercise, to manage the symptoms of PCOS. Lastly, the main goal of PCOS detection research is to make a more accurate and efficient method of identifying and diagnosing PCOS, as well as to find new ways of treating the condition to improve the quality of life of the people with PCOS.

Chapter 2

Detailed Literature Review

This paper [4] has depicted the process of detecting PCOS using Deep Neural Network via image segmentation. In this research, CNN (Convolutional Neural Network) is adopted which can attain 76.36 performance in the testing phase. This can detect the disease more precisely and accurately as it works in layers. At first ultrasound image of two ovaries is given as input in JPG format ,whose intensity and quality is improved via histogram equalization. Then through image enhancement, its brightness level is adjusted. Follicles form is achieved by comparison of two threshold methods : Global Basic and Otsu threshold. After the binarization and image cleaning phase , region based image segmentation is adopted to detect follicles and watershed image segmentation is introduced to separate foreground and background of the image. Finally , the disease is classified based on it's severity into mild, moderate or high level using KNN algorithm,which has an accuracy of 78.81. It's a time saving approach which will be more beneficial to doctors.

This paper[3] describes the identification of Polycystic Ovary Syndrome (PCOS) by analyzing ultrasound images of ovaries using various image processing techniques after collecting basic information about size , position and ovarian follicle count. At first an ultrasound image is transformed into a gray-scale image using RGB-GrayScale conversion. Then the image is divided into a forefront and foundation using image thresholding. After that speckle noise in ultrasound images are undermined with dot clamor using Speckle Noise Elimination. The edge of the follicles is found using the Canny edge detection method. Finally, the SIFT (Scale- Invariant Feature Transform) algorithm is adopted to identify the presence of the syndrome. For training and classification of analyzed data SVM (Support Vector Machine) is adopted as it has highest accuracy of 94.40 , while other two ML algorithms named Naive Bayes and Decision tree have accuracy of 88.25 and 87.54 respectively. Thus, PCOS is detected accurately using above machine learning methods and algorithms.

In this paper[2], researchers want to detect polycystic ovary syndrome (PCOS) by using image segmentation of ultrasound images of ovaries using AI techniques. Firstly, a grayscale colored ultrasound image is given as input. Then, threshold methods Global Basic threshold and Otsu threshold are applied. Background and foreground images are separated. Also Convolution Neural Network (CNN) is used to differentiate the images into PCOS and nonPCOS classes. CNN method, K-means algorithm, Gaussian edge operator has also been used. Using Bayesian and Logistic Regres-

sion (LR) classifiers, vector features are classified. The accuracy of the Bayesian classifier is 93.93 whereas it is 91.04 in LR. In differentiating normal and polycystic ovary, Linear discriminant, K Nearest Neighbour (KNN), Support Vector Machine (SVM) were used with the accuracy of 92.86, 91.43 and 91.43 respectively. It is very tough to recognise this disease through traditional methods but with the help of AI method PCOS can be diagnosed easily with higher accuracy.

The paper[5] represents automated methods for identifying follicles for PCOS diagnosis. Here in this automated method, follicles are detected using image processing techniques which consist of preprocessing of images, segmentation, extracting of features and finally with the help of classifiers. Among which segmentation is very crucial for that reason these methods are being present in this paper with their accuracy rate. Watershed segmentation method is used for image enhancement in pre-processing of ultrasound images. Region growing method is used with a recognition rate of around 78. A modified version of this method is also here with more efficiency. Edge based method is also applied and found to be more efficient than manual detection. Active contour method is another method without edge based that shows less number of false acceptance and false rejections which is indicating the accuracy of the proposed method. The Threshold method has 90 accuracy. C-means and K-means clustering algorithms are also used. Here, K-means performs with a higher accuracy of 96. Using these methods obstructions can be overcome in finding follicles.

The paper [1] describes how determining the number and size of ovarian follicles by manually interpreting ultrasonographic images is currently the only way to diagnose PCOS. However, the variability, reproducibility, and efficiency of this approach may be low. The paper suggests employing an automated system to segment the follicles and identify geometric components quantitatively using stereology and Euclidean distance approaches in order to get over these issues. The suggested approach is designed to help physicians accurately analyze PCO and identify PCOS using ultrasonographic pictures.

In this paper [10] An automated deep learning technique is suggested in this paper for the supplementary identification of PCOS. Scleral alterations are used by the algorithm as potential PCOS diagnostic signals. The method was used on a dataset of 721 Chinese women's full-eye pictures, 388 of whom had PCOS. An enhanced U-Net is used in the algorithm to separate scleral images from full-eye images, and a Resnet model is used to extract deep features. For classification, a multi-instance model is next applied. The algorithm's average AUC is 0.979 and its classification accuracy is 0.929, showing that deep learning has a chance to diagnose PCOS.

In this paper [8] a method is proposed by which Polycystic ovary syndrome (PCOS) can be determined saving doctor's time by using image processing of ultrasound images. This method takes an ultrasound image with gray-scale colored data in JPG format as input. Then the image is enhanced and partitioned using OSTU thresholding. Image binarization is also done then segmentation, features extraction and lastly classification helps to diagnose the disease. This is a detailed diagnosis method using the information of the number of detected follicles, texture and geometric pa-

rameters. This automated method works with higher accuracy with KNN classifiers. Using this method can help to diagnose this disease with up to 97 accuracy.

In this paper [7] using ultrasound images as dataset researchers identified polycystic ovary syndrome with follicle recognition. Important information about the ovary is obtained through ultrasound imaging of the follicles, including the type of cyst, the wide variety of follicles, and the size of the follicles' response to hormonal imbalance. Data gathering is the initial phase in their procedure. Dataset includes USG images with PCOS status along with values for BMI, cycle length, postmenstrual LH, and FSH levels. Patients with conditions like hypercortisolism or thyroid issues are not eligible for this study.To improve the effectiveness of the images obtained, preprocessing techniques like Gray scaling and Histogram equalization are used. Feature extraction, a multi scale morphological approach is used to extract the dark or bright properties from the source image. Then comes the segmentation part by which images are extracted from source images. Using thresholding, edge detection and binarization follicles are detected . In this paper they used SVM ,KNN and Logistic regression and they got accuracy over 90.

This paper [9] is dedicated to detect polycystic ovary syndrome using ovarian ultrasound images through convolutional neural network and pretrained convolutional neural network where a convolutional machine leaning architecture PCONet has been developed using five convolutional layer to detect PCOs with highest precision. Besides, an image recognition model called fine-tuned Inception V3 which contains 42 layers has also been applied for the classification of ultrasound images achieved from different kind of ovaries. In this work, 2 data sets have been use where 1st dataset was dedicated to train and validate the models where training images have been augmented to overcome the image limitations. On the other hand, 2nd dataset has ensured an impartial performance evaluation of the models. Here, both 1st and 2nd dataset included photos of varying dimensions where all the photos have been rescaled to 224 $\times$ 224 pixels. Henceforth, ImageDataGenerator has been used to normalize the image data. Eventually PCONet has been admired since it has proved its superiority and came up with the precision of of 98.12, whereas fine-tuned inception V3 demonstarted the accuracy of 96.56 accuracy.

This paper [6] introduced an automated detection and classification system for Polycystic Ovary Syndrome using machine learning algorithm where ultrasound images have been used to detect PCOS by scrutinizing affected and unaffected cases. According to this paper, multiple ultrasound images differing one from another had been collected and preprocessed using Gaussian low pass filter where images have been equally cropped with the sizes of 256*256. Gaussian low pass filter is a superior technique to annihilate noise since ultrasound images generate strenuous noise. To accomplish the experiment, 90 normal images, 25 cystic images and 35 PCOS cases had been taken where GE LOGIQ ultrasound imaging system has been embedded to get the highest precision. While preprocessing images, ultrasound images get transformed to gray scale from RGB where ROI gets evicted from grey scale. Henceforth, GIST-MDR technique has been used for feature extraction after completing image segmentation through multilevel thresh holding process. This GIST-MDR technique would eventually enhance the accuracy of classification. Thus, this proposed method

came up with the highest precision of 93.82 comparing with other techniques.

Chapter 3

Working Plan

Review of Literature: Conduct a thorough analysis of the literature on the detection of PCOS using ultrasound pictures. Determine the strategies and procedures that are currently most effective in the industry. Image acquisition: Collect a set of ultrasound pictures of the ovaries from PCOS patients as well as a set from healthy controls. Image processing: To extract pertinent features from the ultrasound images, use cutting-edge image processing techniques like morphological analysis and texture analysis. Based on the outcomes of the image processing step, choose the traits that most clearly distinguish PCOS ovaries from healthy ovaries. Model Development: Using the chosen features, create a machine learning model. Then, using the gathered photos, train and test the model. Model Evaluation: Use measures like accuracy, precision, and recall to assess the performance of the created model. Validation: Verify the model using a different dataset and, if necessary, make the necessary changes to enhance performance. Further Research: Investigate more potential uses for the created model while enhancing the precision and efficacy of PCOS identification using ultrasound pictures.

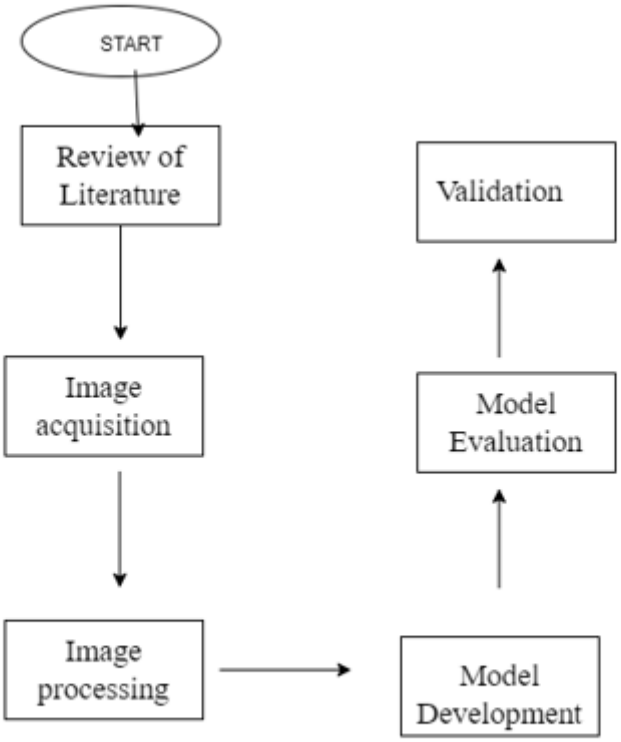


Figure 3.1: Flowchart of work plan

Chapter 4

Conclusion

In conclusion, because ultrasonography is non-invasive and can produce precise images of the ovaries, it is a useful tool for identifying polycystic ovarian syndrome (PCOS). Using cutting-edge image processing methods like morphological analysis and texture analysis, PCOS may be precisely diagnosed and its development tracked. To ensure a thorough and precise diagnosis, ultrasound imaging should be used in conjunction with other diagnostic techniques, such as blood tests and physical examinations. Further study is required to increase the precision and effectiveness of PCOS detection utilizing ultrasound pictures.

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